Test 1 – Grade 9

Name: _____
Grade / Class: 9

Section A: English Language & Literature

1. Reading Comprehension
Read the passage and answer the questions:

"Lina participated in a wildlife conservation camp during her summer break. She learned to track animals, plant native trees, and conduct surveys of local bird species. The experience taught her teamwork, patience, and responsibility for the environment."

a) What activities did Lina perform at the camp?
b) How did Lina feel about the experience?
c) What skills did she develop while participating?
d) Write a four-sentence summary of the passage in your own words.

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2. Grammar & Vocabulary

a) Combine using an appropriate conjunction:
"The storm was approaching. We completed our work on time."

b) Identify all verbs and adverbs in the sentence:
"The students quietly completed their assignments before the deadline."

c) Correct the error:
"Each of the players have practiced for hours."

d) Change to indirect speech:
"The teacher said, 'Tomorrow we will have a science quiz.'"

e) Replace the underlined word with a synonym:
"The view from the mountain was breathtaking."

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3. Writing

Write a short essay (120–150 words) on:
“How Social Media Affects Teenagers”
Include positive and negative effects, and your personal opinion.

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4. Literature Analysis

a) Explain the theme of responsibility in a story you have read recently.
b) Identify the metaphor in this sentence:
"Time is a thief that steals our precious moments."
c) Analyze the character of the main protagonist in your selected novel or story.

Section B: Mathematics

1. Algebra

a) Solve for x:
7x−15=2x+207x - 15 = 2x + 207x−15=2x+20

b) Factorize:
x2−11x+24x^2 - 11x + 24x2−11x+24

c) Simplify:
4p2q−7pq2+3p2q4p^2q - 7pq^2 + 3p^2q4p2q−7pq2+3p2q

d) Solve simultaneously:
2x+3y=122x + 3y = 122x+3y=12, x−y=1x - y = 1x−y=1

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2. Geometry & Mensuration

a) Find the area of a trapezium with bases 12 cm and 8 cm, height 5 cm.
b) A cone has radius 7 cm and height 24 cm. Find its volume.
c) Find the distance between points (3, -2) and (-1, 4) in the coordinate plane.

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3. Trigonometry

a) Calculate sin θ if cos θ = 0.6 and θ is acute.
b) Solve for x in: tan2x+1=sec2xtan^2x + 1 = sec^2xtan2x+1=sec2x
c) Find the height of a tree if its shadow is 15 m and the angle of elevation of the sun is 30°.

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4. Ratio, Proportion & Percentage

a) A jacket costs $120. It is discounted by 20%. Find the selling price.
b) 5 notebooks cost $15. How much do 12 notebooks cost?
c) A population increased from 5,000 to 5,750. Find the percentage increase.

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5. Statistics & Probability

a) Marks of students in an exam: 12, 15, 18, 12, 14, 16, 12. Find mean, median, mode.
b) Draw a frequency table for the data above.
c) A bag contains 5 red, 7 blue, and 8 green balls. Find the probability of picking a red ball.

Section C: Science

1. Biology

a) Name the parts of the human digestive system.
b) Describe the process of photosynthesis and write its chemical equation.
c) Differentiate between xylem and phloem.
d) Explain two adaptations of desert plants.
e) Define ecosystem and give one example.

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2. Physics

a) A car travels 200 m in 10 s. Calculate its average speed.
b) Define kinetic energy and give one example.
c) Explain the difference between conduction and convection of heat.
d) A light ray strikes a plane mirror at 30°. Find the angle of reflection.
e) State Newton's three laws of motion with examples.

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3. Chemistry

a) Classify as element, compound, or mixture: Sodium, CO_2, Sea water.
b) Write a balanced chemical equation for the reaction of zinc with hydrochloric acid.
c) Differentiate between ionic and covalent bonds.
d) Define acid, base, and give one example of each.
e) Explain the law of conservation of mass.

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4. Environmental Science

a) Name two air pollutants and their effects.
b) List three methods to conserve water.
c) Explain global warming and its main causes.
d) Suggest two ways to reduce plastic waste.

Section D: Social Studies

1. History

a) Name two major revolutions of the 18th century and their outcomes.
b) Explain the impact of the printing press on European society.
c) Identify the Allied and Axis powers in World War II.

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2. Geography

a) Define latitude and explain its effect on climate.
b) Name the largest desert in Asia and its location.
c) Discuss two consequences of deforestation.

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3. Civics / Political Science

a) Differentiate between federal and unitary systems of government.
b) List two responsibilities of citizens in a democracy.

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4. Economics

a) Explain the law of supply and demand with examples.
b) Differentiate between fixed and variable costs.
c) Explain the importance of budgeting in personal finance.

Section E: ICT / Computer Studies

1. Computer Basics

a) Define a computer network. Give one type.
b) Name one input and one output device.

2. Applications

a) Which spreadsheet function finds the maximum value?
b) List two uses of presentation software.

3. Programming Basics

a) Define a loop and give one example in Python.
b) Write a Python snippet to print numbers from 1 to 10.

4. Cyber Awareness

a) Define malware and give one method to prevent it.
b) List two ways to protect personal information online.

Section F: Islamiyat

1. Quran & Hadith

a) Name two moral values taught in the Quran.
b) Narrate a Hadith emphasizing honesty.

2. Aqidah / Belief

a) List the five pillars of Iman.
b) Explain the concept of Tawheed.

3. Fiqh / Worship

a) What is the significance of Salah in daily life?
b) Describe the importance of fasting in Ramadan.

4. Islamic Ethics & Morality

a) How does Islam promote honesty in trade?
b) Give two ways Islam encourages environmental protection.

Section G: Arts & Music

1. Visual Arts

a) Explain perspective in drawing.
b) Name three primary and secondary colors.

2. Music

a) Define melody and harmony.
b) Name one string and one percussion instrument.

Section H: Health & Physical Education

1. Physical Fitness

a) Name two exercises for flexibility.
b) List one recreational and one professional sport.

2. Health & Hygiene

a) Why is a balanced diet important for adolescents?
b) Name two practices to prevent communicable diseases.

3. Safety & Environment

a) State one safety rule during outdoor sports.
b) Name one method to prevent home accidents.